

56-Mahbub

# University of Information Technology & Sciences (UITS)

## Faculty of Science and Engineering

### Department of Computer Science & Engineering

#### Program: B.Sc. in CSE

#### Final Term Examination, Autumn 2025

#### Course Title: Data Structures and Algorithms I

#### Course Code: CSE 0613211/211

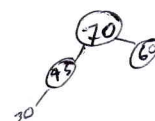
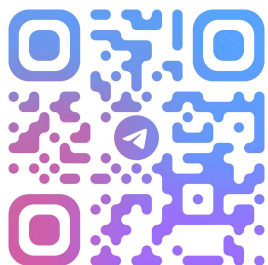
Marks: 50

Time: Three (03) hour

(Answer all questions)

Marks

1. a) You are designing a text editor that allows users to undo their last actions (like deleting or typing text). **Describe** the use of a stack to implement the "Undo" feature? [02]
- b) You have 3 disks on peg A. You need to move them to peg C using peg B. To solve this problem you must follow this conditions: No larger disk may be placed on top of a smaller disk and only one disk may be moved at a time. [04]
  - i. Determine the minimum number of moves required.
  - ii. Identify the first and last disk to be moved.
  - iii. Evaluate the total time needed to solve the 3-disk problem if each move takes 1 second.
- c) Convert the following infix expressions to corresponding postfix expression. [04]
  - i)  $A + B * C - D$
  - ii)  $A + B * (C \wedge D - E) \wedge (F + G * H) - I$
2. a) Consider the following queue of characters, where queue is a circular array which is allocated six memory cells.: [04]  
FRONT=2, REAR=4 QUEUE=\_,A,C,D,\_  
**Describe** the queue (with FRONT and REAR) as the following operations take place.
  - i) F is added to the queue
  - ii) Two letters are deleted.
  - iii) K, L and M are added to the queue
  - iv) N is added to the queue
  - v) Three letters are deleted
  - vi) R is added to the queue
- b) Differentiate between input restricted DEQUEUE and output restricted DEQUEUE. [02]
- c) A small warehouse keeps a digital list of product weights recorded in kilograms at the end of each shift. The manager wants to arrange these weights in ascending order to easily identify the lightest items. Because the system is old and simple, the sorting process works by repeatedly locating the lightest item from the remaining unsorted list and moving it to its correct position. [04]  
**Identify** which sorting algorithm best describes this process. **Explain** why it fits the scenario, and demonstrate how it would sort the list [45, 70, 20, 60, 10, 30].



3. a) Insert the following elements into an empty BST in the given order: 50, 30, 70, 20, 40, 60, 80. [05]

- Draw the final BST.
- For the BST, write the **inorder**, **preorder**, and **postorder** traversals.

- b) Define Hashing. Describe different types of hashing. [03]

- c) Consider the weighted 2-tree in figure 1. Find the external path length and weighted path length. [02]

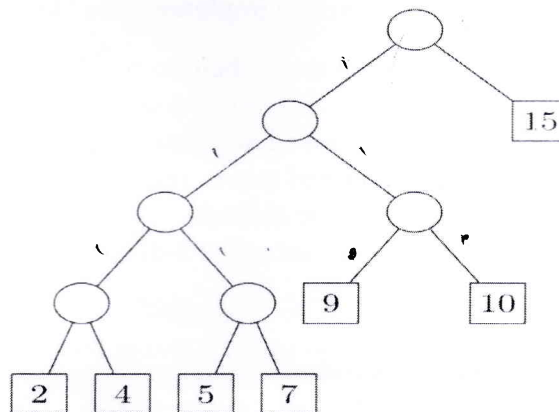


Figure 1: Weighted 2-tree

4. a) Define complete binary tree. You have an empty Max Heap. Apply elements in this order: [25, 40, 35, 20, 50, 10]. [04]

- b) Suppose the following sequences list the nodes of a binary tree T in preorder and inorder, respectively. [04]

Preorder : G, B, Q, A, C, K, F, P, D, E, R, H

Inorder : Q, B, K, C, F, A, G, P, E, D, H, R

Draw the tree based on the above sequence.

- c) Find the adjacency matrix from the graph given in figure 2. [02]

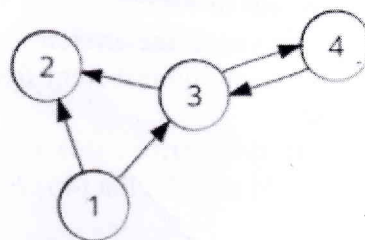
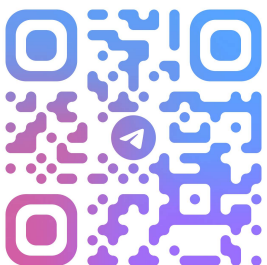


Figure 2: A simple directed graph

50, 30, 70, 20, 40, 60, 80

In: 20  
pre: 50



5. a) Consider the graph in Figure 3. **Apply** Prim's Algorithm and Kruskal Algorithm to generate a minimum spanning tree where the starting node is A. [06]

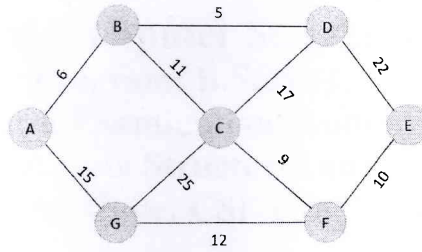


Figure 3: Undirected graph

- b) Given a directed graph represented by an adjacency list: [04]
- 0: [1, 2]
  - 1: [2]
  - 2: [0, 3]
  - 3: [3]
- and the starting node is 0, **print** all nodes reachable from the starting node using DFS.

